Torafirma Systems

ORBIT

Benchmark Results

Public performance report

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1. Summary

On full-vocabulary WikiText-2, ORBIT + readout recorded **236.930** validation perplexity versus **317.103** for the neural-only baseline under the same benchmark split, the same 2,000-step training budget, and a matched approximately 2.27 million parameter budget. This corresponds to a **25.28% lower perplexity** in the recorded comparison.

ORBIT memory-only recorded **240.214** validation perplexity with zero learned memory parameters. The published single-core memory query measurement was **217 ns/token**, compared with **164 us/token** for the small neural forward pass in the same benchmark environment.

Metric	ORBIT	Reference	Recorded difference
Full-vocabulary perplexity	236.930	317.103	25.28% lower
Full-vocabulary memory-only perplexity	240.214	317.103	24.25% lower
5k-vocabulary perplexity	88.183-88.398	109.312	19.23% lower using 2-seed mean
Character-level perplexity	4.090-4.162	5.59	26.19% lower using 2-seed mean
Single-core latency	217 ns/token	164 us/token	~756x lower latency

2. Benchmark conditions

Condition	Full-vocabulary benchmark
Dataset	WikiText-2
Training tokens	2,075,677
Validation tokens	216,347
Vocabulary	33,278 tokens
Training budget	2,000 steps for learned comparison
ORBIT + readout parameters	2,268,356
Neural-only parameters	2,264,256
Parameter difference	0.181%
Latency scope	Single CPU core

3. Full-vocabulary WikiText-2

System	Validation perplexity	Learned parameters	Run count
ORBIT + readout	**236.930**	2,268,356	1
ORBIT memory-only	**240.214**	0 memory parameters	Deterministic
Neural-only baseline	317.103	2,264,256	1

The recorded full-vocabulary ORBIT + readout result is **25.28% lower** in validation perplexity than the matched neural-only reference. The memory-only result is **24.25% lower** than the same reference.

236.930

ORBIT + readout

317.103

Neural-only baseline

25.28% lower validation perplexity

4. 5,000-token WikiText-2

System	Validation perplexity	Learned parameters	Run count
ORBIT + readout	**88.183-88.398**	458,564	2
ORBIT memory-only	**92.934**	0 memory parameters	Deterministic
Neural-only baseline	109.312	454,464	1

The two-run ORBIT + readout mean is **88.291 perplexity**, **19.23% lower** than the neural-only reference. ORBIT memory-only recorded **92.934 perplexity**.

5. Character-level benchmark

System	Validation perplexity	Run count
ORBIT + readout	**4.090-4.162**	2
ORBIT memory-only	**4.201**	Deterministic
Transformer reference	5.59	1

The two-run ORBIT + readout mean is **4.126 perplexity**, **26.19% lower** than the recorded transformer reference.
ORBIT memory-only recorded **4.201 perplexity**.

6. Single-core CPU performance

Measurement	Recorded result
ORBIT memory query	**217 ns/token**
Small neural forward pass	**164 us/token**
Latency ratio	**approximately 756x**
Memory table construction	**1.93 s for 2,075,677 training tokens**

7. Published result statement

> *On full-vocabulary WikiText-2, using the same benchmark split, the same 2,000-step training budget, and a matched approximately 2.27M parameter budget, ORBIT + readout reached **236.9 validation perplexity** versus **317.1** for the neural-only baseline, a **25.3% reduction**. ORBIT memory-only reached **240.2 validation perplexity** with zero learned memory parameters. The published single-core memory query measured **217 ns/token**.*